

# Plotting Higher Order Solution Families (Rachel Batra)

We will plot the family of solutions of the given third order diff eqn:

$$1. y''' - 5y'' + 8y' - 4y = 0$$

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--> eqn : diff ( y ( x ) , x , 3 ) - 5 * diff ( y ( x ) , x , 2 ) + 8 * diff ( y ( x ) , x ) - 4 * y ( x ) = 0 ;
sol : desolve ( eqn , y ( x ) ) ;
sol1 : ev ( sol , y ( 0 ) = c1 , diff ( y ( x ) , x ) = c2 , diff ( y ( x ) , x , 2 ) = c3 ) ;
v1 : ev ( sol1 , c1 = 1 , c2 = 2 , c3 = 3 ) $
v2 : ev ( sol1 , c1 = 2 , c2 = 1 , c3 = 3 ) $
v3 : ev ( sol1 , c1 = 1 , c3 = 2 , c3 = 2 ) $
v4 : ev ( sol1 , c1 = 2 , c2 = 3 , c3 = 1 ) $
v5 : ev ( sol1 , c1 = 3 , c2 = 1 , c3 = 2 ) $
v6 : ev ( sol1 , c1 = 3 , c2 = 2 , c3 = 1 ) $ ,
wxplot2d ( [ rhs ( v1 ) , rhs ( v2 ) , rhs ( v3 ) , rhs ( v4 ) , rhs ( v5 ) , rhs ( v6 ) ] , [ x , 2 , 4 ] , [ legend , "1" , "2" , "3" , "4" , "5" , "6" ] ) $
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$$\frac{d^3}{dx^3}y(x) - 5 \left( \frac{d^2}{dx^2}y(x) \right) + 8 \left( \frac{d}{dx}y(x) \right) - 4y(x) = 0$$

$$y(x) = \%e^x \left( \frac{d^2}{dx^2}y(x) \Big|_{x=0} - 4 \left( \frac{d}{dx}y(x) \Big|_{x=0} \right) + 4y(0) \right) + x \%e^{2x} \left( \frac{d^2}{dx^2}y(x) \Big|_{x=0} \right) + \%e^{2x} \left( - \frac{d^2}{dx^2}y(x) \Big|_{x=0} + 4 \left( \frac{d}{dx}y(x) \Big|_{x=0} \right) - 3y(0) \right) - 3x \%e^{2x} \left( \frac{d}{dx}y(x) \Big|_{x=0} \right)$$

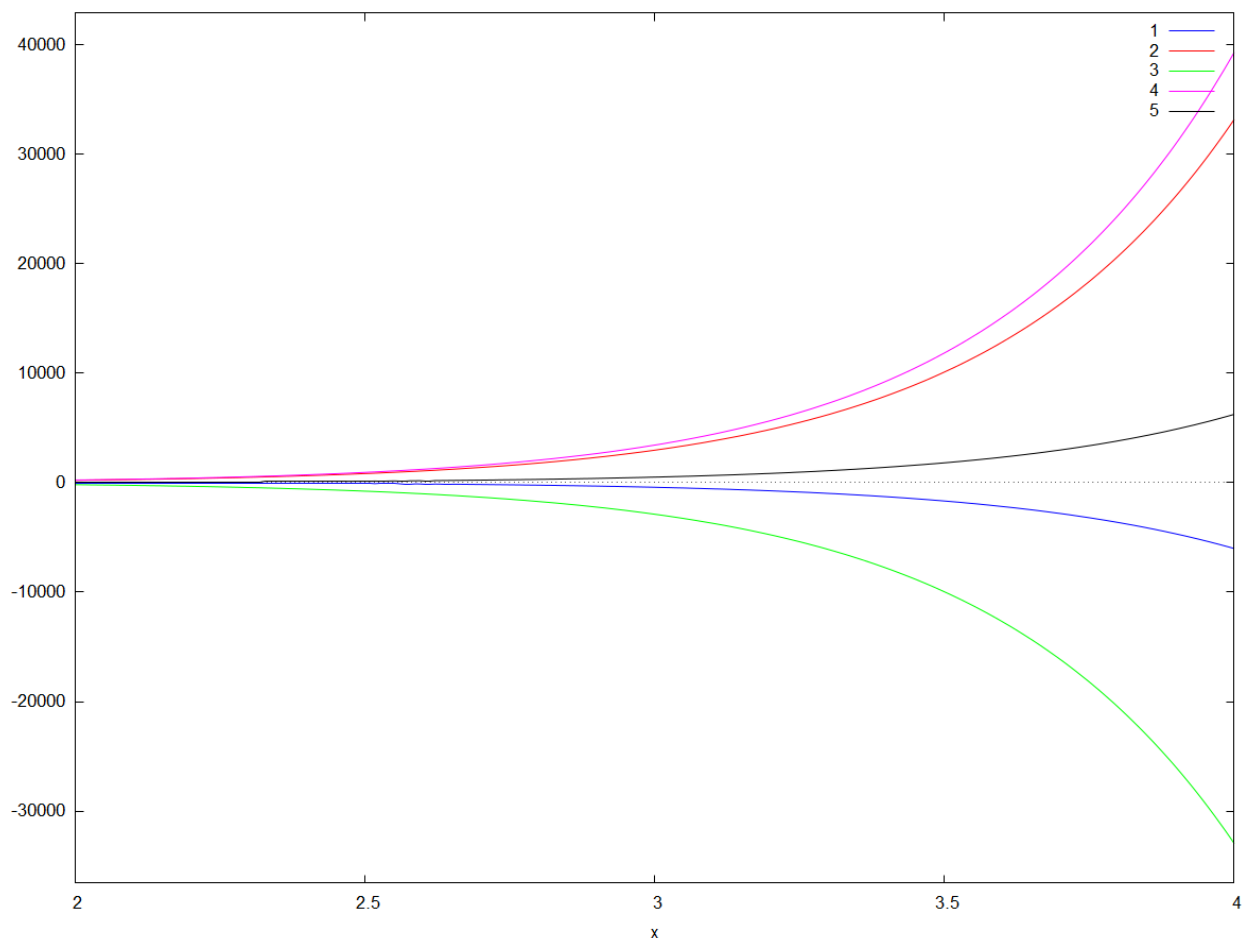
$$y(x) = c3x \%e^{2x} - 3c2x \%e^{2x} + 2c1x \%e^{2x} + (-c3 + 4c2 - 3c1) \%e^{2x} + (c3 - 4c2 + 4c1) \%e^x$$

%errorincorrect syntax: , is not a prefix operator

%error,

%error \textasciicircum

%defaultplot2d: expression evaluates to non-numeric value everywhere in plotting range.



$$2. y''' - 12y'' + 48y' - 64y = 12 - 32\exp(-8x) + 2\exp(4x)$$

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--> eqn : diff ( y ( x ) , x , 3 ) - 12 * diff ( y ( x ) , x , 2 ) + 48 * diff ( y ( x ) , x ) - 64 * y ( x ) = 12 - 32 * \%e ^ ( - 8 * x ) + 2 * \%e ^ ( 4 * x ) ;
sol : desolve ( eqn , y ( x ) ) ;
sol1 : ev ( sol , y ( 0 ) = c1 , diff ( y ( x ) , x ) = c2 , diff ( y ( x ) , x , 2 ) = c3 ) ;
v1 : ev ( sol1 , c1 = 1 , c2 = 2 , c3 = 3 ) $
v2 : ev ( sol1 , c1 = 2 , c2 = 1 , c3 = 3 ) $
v3 : ev ( sol1 , c1 = 1 , c3 = 2 , c3 = 2 ) $
v4 : ev ( sol1 , c1 = 2 , c2 = 3 , c3 = 1 ) $
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d^3
dx^3 y(x) - 12 \left( \frac{d^2}{dx^2} y(x) \right) + 48 \left( \frac{d}{dx} y(x) \right) - 64 y(x) = 2 e^{4x} - 32 e^{-8x} + 12

y(x) = \frac{x^2 e^{4x} \left( 3 \left( \frac{d^2}{dx^2} y(x) \Big|_{x=0} \right) - 24 \left( \frac{d}{dx} y(x) \Big|_{x=0} \right) + 48 y(0) + 1 \right)}{18} + \frac{x^2 e^{4x} \left( \frac{d^2}{dx^2} y(x) \Big|_{x=0} \right)}{3} + \frac{x e^{4x} \left( 36 \left( \frac{d}{dx} y(x) \Big|_{x=0} \right) - 144 y(0) - 19 \right)}{54} - \frac{8 x^2 e^{4x} \left( \frac{d}{dx} y(x) \Big|_{x=0} \right)}{54} + \frac{1}{3}

y(x) = \frac{x^3 e^{4x}}{3} + \frac{(3c3 - 24c2 + 48c1 + 1)x^2 e^{4x}}{18} + \frac{c3 x^2 e^{4x}}{3} - \frac{8c2 x^2 e^{4x}}{3} + \frac{16c1 x^2 e^{4x}}{3} + \frac{x^2 e^{4x}}{9} + \frac{(36c2 - 144c1 - 19)x e^{4x}}{54} + \frac{c2 x e^{4x}}{3} - \frac{4c1 x}{3}

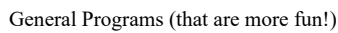
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%error,

%error \textasciicircum

%defaultplot2d: expression evaluates to non-numeric value everywhere in plotting range.

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%defaultfalse
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- 1////
- 2////
- 3////
- 4////
- 5////
- 6////
- 7////
- 8////

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--> s:0;
    for i: 1 while ( i <= 10 ) do
      s: s + i;
      print ( s ) $
    done
55

--> block ( s: 0 , for i: 1 while ( i <= 10 ) do s: s + i , print ( s ) ) $
55

--> block ( s: 1 ,
    for i: 1 while ( i <= 10 ) do
      s: s * i ,
      print ( s ) ) $
3628800

--> block (
  n: read ( "enter the number to find its factorial" ) ,
  s: 1 ,
  for i: 1 while ( i <= n ) do
    s: s * i ,

  print ( "The factorial of" , n , "is" , s ) ) $

enter the number to find its factorial7;
The factorial of7is5040

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